CBSE Class 10 Maths Basic Question Paper 2023

Set 1 (430/5/1) Solutions

SECTION - A

Section - A consists of Multiple-Choice type questions of 1 mark each.

1. The 8th term of an A.P. is 17 and its 14th term is 29 . The common difference of this A.P. is : 1

(a) 3

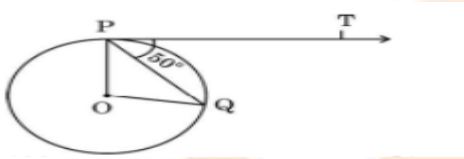
(b) 2

(c) 5

(d) -2

Answer: (b) 2

2. In the given figure, O is the centre of a circle, P Q is a chord and the tangent PT at P makes an angle of 50° with PQ . The measure of $\angle POQ$ is : 1



(a) 130°

(b) 100°

(c) 90°

(d) 75°

Answer: (b) 100°

3. One card is drawn at random from a well shuffled deck of 52 playing cards. What is the probability of getting ' 4 of hearts'? 1

(a) $\frac{1}{52}$

(b) $\frac{1}{13}$

(c) $\frac{1}{26}$

(d) $\frac{1}{6}$

Answer: (b) $\frac{1}{13}$

4. The distance between the points $A(0,6)$ and $B(-6,2)$ is : **1**

(a) 6 units

(b) $2\sqrt{6}$ **units**

(c) $2\sqrt{13}$ **units**

(d) $13\sqrt{2}$ **units**

Answer: (c) $2\sqrt{13}$ **units**

5. The value(s) of k for which the roots of quadratic equation $x^2+4x+k=0$ are real, is : **1**

(a) $k \geq 4$

(b) $k \leq 4$

(c) $k \geq -4$

(d) $k \leq -4$

Answer: (d) $k \leq -4$

6. LCM of $\left(2^3 \times 3 \times 5\right)$ and $\left(2^4 \times 5 \times 7\right)$ is: **1**

(a) 40

(b) 560

(c) 1680

(d) 1120

Answer: (d) 1120

7. If one zero of the quadratic polynomial $kx^2 + 3x + k$ is 2 , then the value of k is: **1**

(a) $-\frac{6}{5}$

(b) $\frac{6}{5}$

(c) $\frac{5}{6}$

(d) $-\frac{5}{6}$

Answer: (c) $\frac{5}{6}$

8. If the lines represented by equations $3x+2my=2$ and $2x+5y+1=0$ are parallel, then the value of m is: 1

(a) $\frac{2}{5}$

(b) $-\frac{5}{4}$

(c) $\frac{3}{2}$

(d) $\frac{15}{4}$

Answer: (b) $-\frac{5}{4}$

9. $\triangle ABC \sim \Delta DEF$ **and their perimeters are 32 cm and 24 cm respectively. If** $AB = 10\,cm$**, then DE equals :** **1**

(a) 8 cm

(b) 7.5 cm

(c) 15 cm

(d) $5\sqrt{3}\,cm$

Answer: (b) 7.5 cm

10. The two roots of the equation $3x^2 - 2\sqrt{6}x + 2 = 0$ are:

(a) real and distinct

(b) not real

(c) real and equal

(d) rational

Answer: (a) real and distinct

11. If $\sin\theta = \frac{a}{b}$, then $\sec\theta$ is equal to $\left(0 \le \theta \le 90^\circ\right)$: **1**

(a) $\frac{a}{\sqrt{b^2 - a^2}}$

(b) $\frac{b}{\sqrt{b^2 - a^2}}$

(c) $\frac{\sqrt{b^2 - a^2}}{b}$

(d) $\frac{\sqrt{b^2 - a^2}}{a}$

Answer: (d) $\frac{\sqrt{b^2 - a^2}}{a}$

12. The sum of the first 100 even natural numbers is: **1**

(a) 10100

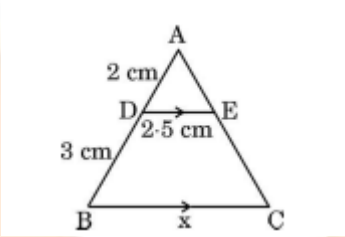
(b) 2550

(c) 5050

(d) 10010

Answer: (b) 2550

13. In the given figure, $AD = 2\,cm, DB = 3\,cm, DE = 2.5\,cm$ **and** $DE \parallel BC$**. The value of x** **1**



(a) 6 cm

(b) 3.75 cm

(c) 6.25 cm

(d) 7.5 cm

Answer:(d) 7.5 cm

14. A circle is of radius 3 cm. The distance between two of its parallel tangents is : 1

(a) 12 cm

(b) 6 cm

(c) 3 cm

(d) 4·5 cm

Answer: (b) 6 cm

15. The median class for the data given below is. 1

Class	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120
Frequency	10	12	14	13	17

(a) 80-100

(*b*) **20-40**

(*c*) **40-60**

(*d*) **60-80**

Answer: (*d*) 60-80

16. If $\sin\theta = \frac{3}{4}$, then $\frac{(\sec^2\theta - 1)\cos^2\theta}{\sin\theta}$ equals : **1**

(a) $\frac{3}{5}$

(b) $\frac{3}{4}$

(c) $\frac{4}{3}$

(d) $\frac{9}{16}$

Answer:(a) $\frac{3}{5}$

17. In two triangles $\triangle PQR$ and ΔABC, it is given that $\frac{AB}{BC} = \frac{PQ}{PR}$. For these two triangles to be similar, which of the following should be true? **1**

(a) $\angle A = \angle P$

(b) $\angle B = \angle Q$

(c) $\angle B = \angle P$

(d) $CA = QR$

Answer: (a) $\angle A = \angle P$

18. Mean and median of some data are 32 and 30 respectively. Using empirical relation, mode of the data is : 1

(a) 36

(b) 26

(c) 30

(d) 20

Answer: (c) 30

Questions number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

19. Assertion (A) : When two coins are tossed together, the probability of getting no tail is $\frac{1}{4}$. **1**

Reason (R) : The probability $P(E)$ of an event E satisfies $0 \le P(E) \le 1$.

Answer: (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

20. Assertion (A) : The surface area of largest sphere that can be inscribed in a hollow cube of side ' a ' cm is $\pi a^2 \, cm^2$.

Reason (R) : The surface area of a sphere of radius ' r' is $\frac{4}{3}\pi r^3$. **1**

Answer: (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

21. Find LCM of 576 and 512 by prime factorization. **2**

Solution:

Prime factorization of 576:

$576 = 2^6 \times 3^2$

Prime factorization of 512:

$512 = 2^9$

LCM is found by taking the highest power of all prime factors:

$\text{LCM} = 2^9 \times 3^2 = 512 \times 9 = 4608$

Answer: The LCM of 576 and 512 is 4608.

22.(a) Evaluate: $\dfrac{\sin 30^\circ + \tan 45^\circ}{\sec 30^\circ + \cot 45^\circ}$ **2**

Solution:

Substitute the trigonometric values:

$\sin 30^\circ = \dfrac{1}{2}, \quad \tan 45^\circ = 1, \quad \sec 30^\circ = \dfrac{2}{\sqrt{3}}, \quad \cot 45^\circ = 1$

Therefore, the expression becomes:

$$\frac{\frac{1}{2}+1}{\frac{2}{\sqrt{3}}+1} = \frac{\frac{3}{2}}{\frac{2+\sqrt{3}}{\sqrt{3}}} = \frac{3\sqrt{3}}{2(2+\sqrt{3})}$$

Answer: Simplified value of the expression is $\dfrac{3\sqrt{3}}{2(2+\sqrt{3})}$.

OR

(b) For $A = 30^\circ$ and $B = 60^\circ$, verify that: $\sin(A+B) = \sin A \cos B + \cos A \sin B$ **2**

Solution:

The left-hand side:

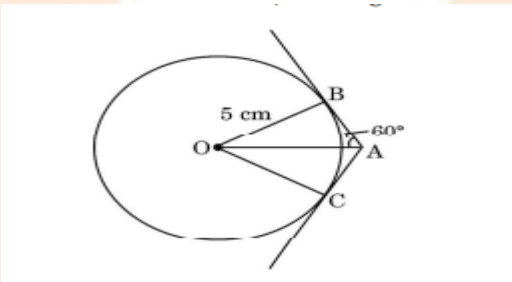
$\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$

The right-hand side:

$\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ = \frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{1}{4} + \frac{3}{4} = 1$

Answer: Since LHS = RHS, the identity is verified.

23. In the given figure, tangents AB and AC are drawn to a circle centred at O. If $\angle OAB = 60^\circ$ and $OB = 5$ cm, find lengths OA and AC **2**



Solution: *Since* AB and AC are tangents to the circle, $\angle OAB = \angle OAC = 90^\circ$ (tangent to a circle is perpendicular to the radius at the point of contact).

In $\triangle OAB$, $\angle OAB = 60^\circ$, $\angle OBA = 90^\circ$, and $\angle BOA = 30^\circ$.

Using trigonometric ratios in $\triangle OAB$:

$$\cos 60^\circ = \frac{OA}{OB} = \frac{1}{2} \quad \Rightarrow \quad OA = OB \times \frac{1}{2} = 5 \times \frac{1}{2} = 2.5 \text{ cm}$$

The length of tangent AB $(or$ AC) is given by:

$$AB = \sqrt{OB^2 - OA^2} = \sqrt{5^2 - 2.5^2} = \sqrt{25 - 6.25} = \sqrt{18.75} = 4.33 \text{ cm}$$

Answer: $OA = 2.5$ cm, $AB = AC = 4.33$ cm.

24. (a) Show that $A(1,2)$, $B(5,4)$, $C(3,8)$, and $D(-1,6)$ are vertices of a parallelogram ABCD. 2

Solution:

Find the midpoint of diagonals AC and BD $.Midpointof$ AC:

$$\text{Midpoint} = \left(\frac{1+3}{2}, \frac{2+8}{2}\right) = (2,5)$$

Midpoint of BD:

$$\text{Midpoint} = \left(\frac{5-1}{2}, \frac{4+6}{2}\right) = (2,5)$$

Since both midpoints are the same, AC and BD bisect each other, proving that ABCD $isaparallelogram.item\,size$

Or

24. B) Show that the Points A(3, 0), $B(6,4)$, and $C(-1,3)$ are vertices of a right-angled triangle. **2**

Solution:

Calculate the lengths of sides AB, BC, and CA using the distance formula:

$$AB = \sqrt{(6-3)^2 + (4-0)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} = 5$$

$$BC = \sqrt{(6+1)^2 + (4-3)^2} = \sqrt{7^2 + 1^2} = \sqrt{49+1} = \sqrt{50}$$

$$CA = \sqrt{(3+1)^2 + (3-0)^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

Check if

$$AB^2 + AC^2 = BC^2 \; 5^2 + 5^2 = 50 \quad \text{and} \quad \sqrt{50}^2 = 50$$

Since the condition satisfies, $\triangle ABC$ is a right-angled triangle.

25. Find the sum of the first 15 terms of the A.P.: $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}, \ldots$ **2**

Solution:

The first term $a = \frac{1}{15}$.

Common difference $d = \frac{1}{12} - \frac{1}{15} = \frac{5-4}{60} = \frac{1}{60}$.

The sum of the first n terms is:

$$S_n = \frac{n}{2} \times [2a + (n-1)d]$$

Substituting $a=\frac{1}{15}$, $d=\frac{1}{60}$, and $n=15$:

$$S_{15}=\frac{15}{2}\times\left[\frac{2}{15}+14\times\frac{1}{60}\right]=\frac{15}{2}\times\left[\frac{2}{15}+\frac{14}{60}\right]=\frac{15}{2}\times\left[\frac{2}{15}+\frac{7}{30}\right]=2.5$$

Answer: The sum of the first 15 terms is 2.5.

SECTION C

This section comprises short answer (SA) type questions of 3 marks each.

26. (a) Sabina went to a bank ATM to withdraw ₹2000. She received ₹50 and ₹100 notes only. If Sabina got 25 notes in all, how many notes of ₹50 and ₹100 did she receive? 3

Solution:

Let the number of ₹50 notes be x and the number of ₹100 notes be y.

The total number of notes is 25: $x+y=25$

The total amount is ₹2000: $50x+100y=2000$

Solving the equations:

$x + y = 25$

$50x + 100(25 - x) = 2000 \quad \Rightarrow \quad 50x + 2500 - 100x = 2000 \quad \Rightarrow \quad -50x = -500 \quad \Rightarrow \quad x = 10$

Substituting $x = 10$ into the first equation:

$10 + y = 25 \quad \Rightarrow \quad y = 15$

Answer: Sabina received 10 notes of ₹50 and 15 notes of ₹100.

OR

(b) Five years ago, Amit was thrice as old as Baljeet. Ten years hence, Amit shall be twice as old as Baljeet. What are their present ages? 3

Solution:

Let Amit's present age be x years and Baljeet's present age be y years.

Five years ago: $x - 5 = 3(y - 5) \quad \Rightarrow \quad x - 5 = 3y - 15 \quad \Rightarrow \quad x - 3y = -10 \quad (1)$

Ten years hence: $x + 10 = 2(y + 10) \quad \Rightarrow \quad x + 10 = 2y + 20 \quad \Rightarrow \quad x - 2y = 10 \quad (2)$

Solving equations (1) and (2):

$x - 3y = -10$

$x - 2y = 10 \quad \Rightarrow \quad y = 20$

Substituting $y = 20$ in equation (1):

x - 60 = -10 $\quad \Rightarrow \quad x = 50$

Amit's present age is 50 years, and Baljeet's present age is 20 years.

27.Prove that $(7-2\sqrt{3})$ is an irrational number, given that $\sqrt{3}$ is an irrational number. **3**

Solution:

Assume $(7-2\sqrt{3})$ is rational. Let it be equal to $\frac{a}{b}$ where a and b are integers.

Then: $7-2\sqrt{3}=\frac{a}{b}$

Rearrange:

$2\sqrt{3}=7-\frac{a}{b}$

Since $\sqrt{3}$ is irrational, multiplying it by a rational number 2 still results in an irrational number. However, $7-\frac{a}{b}$ is rational. This contradiction implies that our assumption is wrong.

Therefore, $(7-2\sqrt{3})$ must be irrational.

28.Find the mean of the following data: **3**

Class	0 – 15	15 – 30	30 – 45	45 – 60	60 – 75	75 – 90
Frequency	12	15	11	20	16	6

Solution:

Find the midpoints (x_i) for each class interval: $\text{Midpoints} = 7.5, 22.5, 37.5, 52.5, 67.5, 82.5$

Multiply midpoints by their corresponding frequencies:
$f_i \times x_i = 90, 337.5, 412.5, 1050, 1080, 495$

Calculate the sum of frequencies ($\Sigma f_i = 80$) and sum of $f_i x_i$: $\Sigma f_i x_i = 3465$

Mean:

$$\text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{3465}{80} = 43.31$$

Answer: The mean of the data is 43.31.

29. (a) Determine the ratio in which the point $P(a,-2)$ divides the line segment joining the points $A(-4,3)$ and $B(2,-4)$. Also, find the value of a. **3**

Solution:

Let the ratio be k:1.

The coordinates of the point P dividing AB in the ratio k:1 are given by:

$$P\left(\frac{k\times 2+1\times(-4)}{k+1},\frac{k\times(-4)+1\times 3}{k+1}\right)$$

Since the y-coordinate of P is -2:

$$\frac{-4k+3}{k+1}=-2$$

Solving for k:

$$-4k+3=-2k-2 \quad\Rightarrow\quad -2k=-5 \quad\Rightarrow\quad k=\frac{5}{2}$$

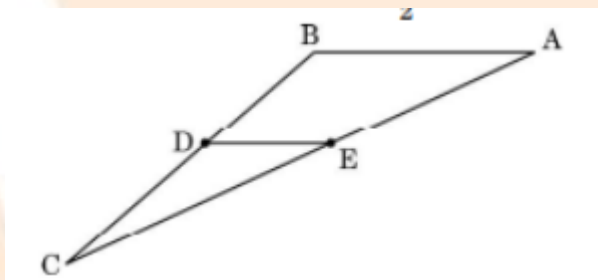
Now, substitute k into the x-coordinate equation:

$$a=\frac{\frac{5}{2}\times 2+(-4)}{\frac{5}{2}+1}=\frac{5-4}{\frac{7}{2}}=\frac{1\times 2}{7}=\frac{2}{7}$$

Answer: The ratio is 5:2, and the value of a is $\frac{2}{7}$.

OR

(b) In the given figure, in $\triangle ABC$, points D and E are midpoints of sides BC and AC respectively. If given vertices are $A(4,-2)$, $B(2,-2)$, and $C(-6,-7)$, then verify the result $DE = \frac{1}{2}AB$. **3**



Solution:

Calculate the coordinates of D and E:

$$D = \left(\frac{2-6}{2}, \frac{-2-7}{2}\right) = (-2, -4.5)$$

$$E = \left(\frac{4-6}{2}, \frac{-2-7}{2}\right) = (-1, -4.5)$$

Calculate the lengths of AB and DE:

$$AB = \sqrt{(4-2)^2 + (-2+2)^2} = \sqrt{2^2 + 0} = 2 \text{ units}$$

$$DE = \sqrt{(-2+1)^2 + (-4.5+4.5)^2} = \sqrt{(-1)^2 + 0} = 1 \text{ unit}$$

Since $DE = \frac{1}{2}AB$, the relationship is verified.

Answer: The relationship $DE = \frac{1}{2}AB$ is verified.

30. Prove that $(\csc A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$

Solution:

Start by expressing $\csc A$, $\sec A$, and $\cot A$ in terms of $\sin A$ and $\cos A$:

$$\csc A = \frac{1}{\sin A}, \quad \sec A = \frac{1}{\cos A}, \quad \cot A = \frac{\cos A}{\sin A}$$

The left-hand side can be expanded and simplified:

$$(\frac{1}{\sin A} - \sin A)(\frac{1}{\cos A} - \cos A)$$

Multiply and simplify:

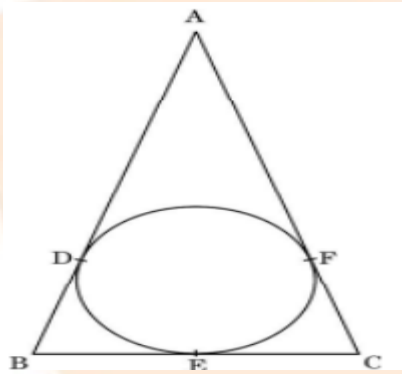
$$= \frac{1-\sin^2 A}{\sin A} \times \frac{1-\cos^2 A}{\cos A}$$

Apply trigonometric identity $1-\sin^2 A = \cos^2 A$ and $1-\cos^2 A = \sin^2 A$:

$$= \frac{\cos^2 A}{\sin A} \times \frac{\sin^2 A}{\cos A} = \frac{\cos A \times \sin A}{\sin A \times \cos A} = \frac{1}{\tan A + \cot A}$$

Answer: The identity is proved.

31.ABC is an isosceles triangle with $AB = AC$, circumscribed about a circle. Prove that BC is bisected at E. **3**



Solution:

Since $AB = AC$ and $AD = AF$ (equal tangents drawn from a common external point), we have:

$BE = BD = CE = CF$

In $\triangle ABC$, since $AB = AC$, the incenter I of $\triangle ABC$ lies on the angle bisector of $\angle BAC$.

Since $AB = AC$ and the incenter I lies on the angle bisector of $\angle BAC$, $BE = EC$. Thus, BC is bisected at E.

Answer: BC is bisected at E.

SECTION D

This section comprises long answer (LA) type questions of 5 marks each.

32. A person walking 48 m towards a tower in a horizontal line through its base observes that the angle of elevation of the top of the tower changes from 45° to 60°. Find the height of the tower and distance of the person, now, from the tower. (Use $\sqrt{3}=1.732$)

Solution:

Let the height of the tower be h meters and the distance of the person from the base of the tower when the angle of elevation is 45° be d meters.

Using the tangent function:

$$\tan 45^\circ = \frac{h}{d} \quad \Rightarrow \quad 1 = \frac{h}{d} \quad \Rightarrow \quad h = d$$

Let the distance of the person from the base of the tower when the angle of elevation is 60° be x meters.

Using the tangent function:

$$\tan 60^\circ = \frac{h}{x} \quad \Rightarrow \quad \sqrt{3} = \frac{d}{x} \quad \Rightarrow \quad x = \frac{d}{\sqrt{3}} = \frac{d}{1.732}$$

The person walks 48 m towards the tower, so

$$d - x = 48 \quad \Rightarrow \quad d - \frac{d}{1.732} = 48$$

Simplifying:

$$d\left(1-\frac{1}{1.732}\right)=48 \quad \Rightarrow \quad d\times\frac{0.732}{1.732}=48 \quad \Rightarrow \quad d\approx 113.56 \text{ meters}$$

Since $h=d$:

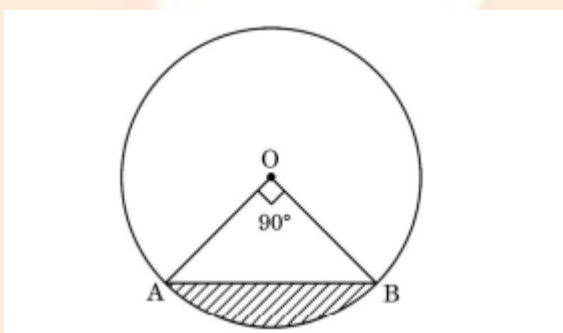
$$h\approx 113.56 \text{ meters}$$

The distance of the person now from the tower:

$$x=\frac{113.56}{1.732}\approx 65.57 \text{ meters}$$

Answer: The height of the tower is approximately 113.56 meters, and the distance of the person from the tower is approximately 65.57 meters.

33. (a) In the given figure, AB is a chord of a circle of radius 7 cm and centered at O. Find the area of the shaded region if $\angle AOB=90^{\circ}$. Also, find the length of minor arc AB.



Solution:

The area of the sector OAB is:

$$\text{Area of sector } OAB=\frac{\theta}{360^{\circ}}\times\pi r^2=\frac{90^{\circ}}{360^{\circ}}\times\frac{22}{7}\times 7^2=\frac{1}{4}\times 154=38.5 \text{ cm}^2$$

The area of the triangle OAB:

$$\text{Area of } \triangle OAB = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \times 7 = 24.5 \text{ cm}^2$$

The area of the shaded region:

$$\text{Area of shaded region} = \text{Area of sector } OAB - \text{Area of } \triangle OAB = 38.5 - 24.5 = 14 \text{ cm}^2$$

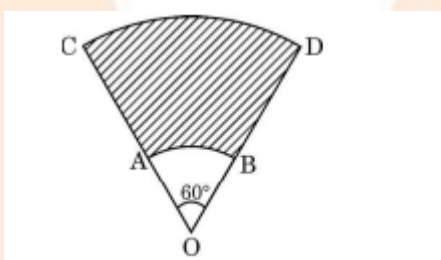
The length of the minor arc AB:

$$\text{Length of arc } AB = \frac{\theta}{360^\circ} \times 2\pi r = \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 7 = \frac{1}{4} \times 44 = 11 \text{ cm}$$

Answer: The area of the shaded region is 14 cm², and the length of the minor arc AB is11cm.

OR

(b) AB and CD are arcs of two concentric circles of radii 3.5 cm and 10.5 cm respectively and centered at O. Find the area of the shaded region if $\angle AOB = 60^\circ$. Also, find the length of arc CD.



Solution: The area of the sector OAB with radius 10.5 cm.

$$\text{Area of sector } OAB = \frac{\theta}{360^\circ} \times \pi R^2 = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 10.5^2 = \frac{1}{6} \times \frac{22}{7} \times 110.25 = 57.75 \text{ cm}^2$$

The area of the sector OCD with radius 3.5 cm:

Area of sector $OCD = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 3.5^2 = \frac{1}{6} \times \frac{22}{7} \times 12.25 = 7.7 \text{ cm}^2$

The area of the shaded region:

Area of shaded region = Area of sector OAB – Area of sector $OCD = 57.75 - 7.7 = 50.05 \text{ cm}^2$

The length of the arc CD:

Length of arc $CD = \frac{\theta}{360^\circ} \times 2\pi R = \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 10.5 = \frac{1}{6} \times 66 = 11 \text{ cm}$

Answer: The area of the shaded region is 50.05 cm², and the length of the arc CD is 11 cm.

34.If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, then prove that the other two sides are divided in the same ratio.

Solution:

Let $\triangle ABC$ have a line DE parallel to BC, intersecting AB at D and AC at E.

By the Basic Proportionality Theorem (Thales' Theorem), if a line is drawn parallel to one side of a triangle to intersect the other two sides, then the line divides those sides in the same ratio.

Therefore, $\frac{AD}{DB} = \frac{AE}{EC}$.

This proves that the sides are divided in the same ratio.

35.

(a) The difference of two numbers is 5 and the difference of their reciprocals is $\frac{1}{10}$. Find the numbers.

Solution:

Let the two numbers be x and y.

Given: $x - y = 5$

$$\frac{1}{y} - \frac{1}{x} = \frac{1}{10} \quad \Rightarrow \quad \frac{x-y}{xy} = \frac{1}{10}$$

Substitute $x - y = 5$:

$$\frac{5}{xy} = \frac{1}{10} \quad \Rightarrow \quad xy = 50$$

Solve the quadratic equation:

$$t^2 - 5t - 50 = 0$$

$$t = \frac{5 \pm \sqrt{25+200}}{2} = \frac{5 \pm \sqrt{225}}{2} = \frac{5 \pm 15}{2}$$

The solutions are $t = 10$ or $t = -5$.

Hence, the numbers are 10 and 5.

OR

(b) Find all the values of k for which the quadratic equation $2x^2 + kx + 8 = 0$ has equal roots. Also, find the roots.

Solution:

For equal roots, the discriminant must be zero:

$b^2 - 4ac = 0$

Here, $a = 2$, $b = k$, and $c = 8$:

$k^2 - 4 \times 2 \times 8 = 0 \quad \Rightarrow \quad k^2 - 64 = 0 \quad \Rightarrow \quad k = \pm 8$

For $k = 8$:

$2x^2 + 8x + 8 = 0 \quad \Rightarrow \quad (x+2)^2 = 0 \quad \Rightarrow \quad x = -2$

For $k = -8$:

$2x^2 - 8x + 8 = 0 \quad \Rightarrow \quad (x-2)^2 = 0 \quad \Rightarrow \quad x = 2$

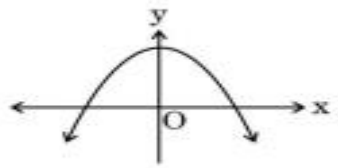
Answer: The values of k are 8 and -8, and the roots are -2 and 2 respectively.

SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study - 1

36. Rainbow is an arch of colours that is visible in the sky after rain or when water droplets are present in the atmosphere. The colours of the rainbow are generally red, orange, yellow, green, blue, indigo, and violet. Each colour of the rainbow makes a parabola. We know that any quadratic polynomial $p(x) = ax^2 + bx + c$ (where $a \geq 0$) represents a parabola on the graph paper.



Answer the following questions:

Based on the above, answer the following questions:

(i)The graph of a rainbow y = f(x) is shown in the figure. Write the number of zeroes of the curve. 1

Solution: The number of zeroes of the curve is equal to the number of times the parabola intersects the x-axis. Here, it intersects at two points, so the number of zeroes is

2.

(ii)If the graph of a rainbow does not intersect the x-axis but intersects the y-axis at one point, then how many zeroes will it have?

Solution: If the graph does not intersect the x-axis but intersects the y-axis at one point, then the parabola has no real zeroes (zero real roots), but the parabola still has complex roots (2 zeroes in the complex number system).

(iii) (a)If a rainbow is represented by the quadratic polynomial p(x) = $x^2+(a+1)x+b$**, whose zeroes are 2 and -3, find the value of a and b. 2**

Solution:

The sum of the zeroes:

$$2+(-3)=a+1 \quad \Rightarrow \quad a+1=-1 \quad \Rightarrow \quad a=-2$$

The product of the zeroes:

$$2\times(-3)=b \quad \Rightarrow \quad b=-6$$

Answer: a = -2, b = -6.

OR

(iii) (b)The polynomial x^2 - 2x - (7p + 3) represents a rainbow. If -4 is a zero of it, find the value of p. 2

Solution:

Substitute x = -4 in the polynomial:

$$(-4)^2-2(-4)-(7p+3)=0 \quad \Rightarrow \quad 16+8-(7p+3)=0$$

Simplify and solve for p:

$$24-3=7p \quad \Rightarrow \quad 21=7p \quad \Rightarrow \quad p=3$$

Answer: p = 3.

Case Study - 2

37. Singing bowls (hemispherical in shape) are commonly used in sound healing practices. Mallet (cylindrical in shape) is used to strike the bowl in a sequence to produce sound and vibration.

One such bowl is shown here whose dimensions are:

Hemispherical bowl has outer radius 6 cm and inner radius 5 cm.

Mallet has height of 10 cm and radius 2 cm.

Based on the above, answer the following questions:

(i)What is the volume of the material used in making the mallet? 1

Solution:

The volume of a cylinder (mallet) is given by:

$V = \pi r^2 h = \pi \times 2^2 \times 10 = 40\pi$

Answer: The volume of the material used in the mallet is 40pi cubic cm.

(ii)The bowl is to be polished from inside. Find the inner surface area of the bowl. 1

Solution:

The inner surface area of a hemisphere is given by:

$A = 2\pi r^2 = 2 \times \pi \times 5^2 = 50\pi$

Answer: The inner surface area of the bowl is $50\pi squarecm$.

(iii) (a)Find the volume of metal used to make the bowl. 2

Solution:

The volume of the material used in the bowl is the difference in the volumes of the outer and inner hemispheres.

Answer: The volume of metal used is $\frac{182\pi}{3}$ cubic cm.

OR

(iii) (b)Find the total surface area of the mallet. (Use π = 3.14) **2**

Solution:

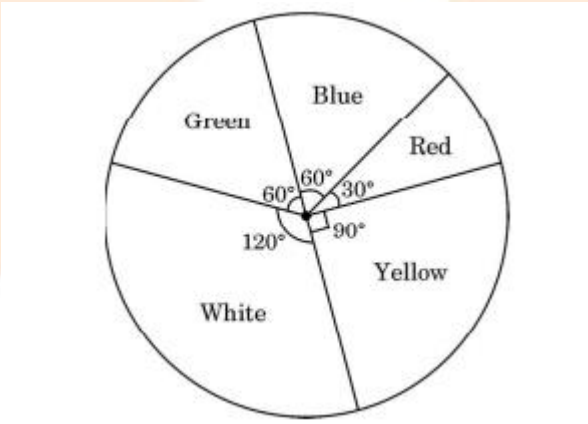
The total surface area of the mallet (cylinder) is:

$$A = 2\pi r(r+h) = 2\times 3.14\times 2\times (2+10) = 2\times 3.14\times 2\times 12 = 150.72 \text{squarecm}$$

Answer: The total surface area of the mallet is 150.72 square cm.

Study - 3

38.Some students were asked to list their favourite colour. The measure of each colour is shown by the central angle of a pie chart given below:



Study the pie chart and answer the following questions:

(i)If a student is chosen at random, then find the probability of his/her favourite colour being white? 1

Solution:

The central angle for white is 120°. The total angle in a pie chart is 360°.

$$P(\text{White})=\frac{120^\circ}{360^\circ}=\frac{1}{3}$$

Answer: The probability is $\frac{1}{3}$.

(ii)What is the probability of his/her favourite colour being blue or green? 1

Solution:

The central angles for blue and green are 60^circ and 60^circ respectively.

$$P(\text{BlueorGreen})=\frac{60^\circ+60^\circ}{360^\circ}=\frac{120^\circ}{360^\circ}=\frac{1}{3}$$

.

Answer: The probability is $\frac{1}{3}$.

(iii) (a)If 15 students liked the colour yellow, how many students participated in the survey? 2

Solution:

The central angle for yellow is 90°. If 15 students correspond to 90°, then:

$$\frac{90^\circ}{360^\circ}=\frac{1}{4}$$

Therefore:

$$\frac{1}{4}\times \text{TotalStudents}=15 \quad \Rightarrow \quad \text{TotalStudents}=15\times 4=60$$

Answer: 60 students participated in the survey.

OR

(iii) (b)What is the probability of the favourite colour being red or blue? 2

Solution:

The central angles for red and blue are 30^circ and 60^circ respectively.

$$P(\text{RedorBlue})=\frac{30^\circ+60^\circ}{360^\circ}=\frac{90^\circ}{360^\circ}=\frac{1}{4}$$

Answer: The probability is $\frac{1}{4}$.